

Abstract. Face Anti-Spoofing (FAS) plays a critical role in securing biometric authentication systems against presentation attacks such as printed photographs, display attacks, and replay videos. This paper proposes a lightweight and edge-oriented FAS framework based on a six-channel MobileNetV3-Small architecture for real-time deployment on Raspberry Pi 4 devices. The proposed method integrates spatial RGB information with temporal difference maps, enabling the network to jointly learn appearance and motion cues without requiring computationally intensive optical flow estimation or 3D reconstruction. To compensate for the lack of temporal information in the LCC-FASD dataset, a dynamic pair generation strategy is introduced to synthesize realistic motion patterns for bona fide and spoof samples. The model further employs transfer-learning initialization and class-weighted focal loss to improve robustness against difficult spoofing attacks. Experimental results demonstrate strong performance with an APCER of 0.81%, BPCER of 1.52%, and ACER of 1.17%, outperforming existing lightweight MobileNetV3-based approaches on the same benchmark. Deployment analysis indicates an estimated inference speed of approximately 8 fps on Raspberry Pi 4, demonstrating the practicality of the proposed framework for embedded biometric authentication systems.

Keywords: Face Anti-Spoofing (FAS), Presentation Attack Detection, MobileNetV3-Small, Edge Computing Raspberry Pi 4

Empowering IoT Security: A Paradigm Shift in Real-Time Face Anti-Spoofing

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1 Introduction

The widespread adoption of facial recognition technologies in biometric authentication systems has significantly strengthened security mechanisms across numerous application domains. However, these systems remain highly vulnerable to Presentation Attacks (PAs), including printed photographs, replay attacks, and three-dimensional mask spoofing. As a result, Face Anti-Spoofing (FAS) has become a critical component for preserving the reliability and security of modern biometric authentication systems. Recent advances in deep learning have substantially improved FAS performance, achieving remarkable detection accuracy across a variety of benchmark datasets. Nevertheless, most state-of-the-art approaches depend on computationally intensive architectures with large memory footprints, limiting their applicability on resource-constrained edge devices such as the Raspberry Pi 4. To address this challenge, research efforts have increasingly focused on developing lightweight and computationally efficient FAS models.

Previous studies have demonstrated the potential of compact architectures for edge-oriented FAS applications. Silva et al. [8] systematically evaluated several lightweight neural networks and reported that the MobileNet family provides an effective balance between detection accuracy and computational efficiency. The feasibility of deploying lightweight biometric security systems on embedded hardware was further demonstrated by Rahman et al. [7], who showed that optimized convolutional neural networks can achieve practical real-time performance on platforms such as the Raspberry Pi 4. Furthermore, Nikitin and Konushin [5] demonstrated that knowledge distillation and structural optimization techniques enable MobileNetV3 architectures to achieve competitive performance while satisfying strict latency requirements in mobile environments.

Despite these advancements, many lightweight FAS solutions rely primarily on static spatial representations extracted from individual RGB

frames. Such approaches often struggle to detect sophisticated presentation attacks that appear realistic. To overcome this limitation, temporal information has been introduced as an additional source of discriminative features. Boulkenafet *et al.* [2] demonstrated that temporal difference maps generated from consecutive video frames can effectively capture subtle physiological cues, including natural head movements and facial muscle dynamics, which are typically absent in spoofing artifacts.

However, a critical examination of recent edge-deployed FAS systems reveals that the full potential of spatio-temporal information remains largely underutilized. Existing approaches often sacrifice either temporal awareness or computational efficiency. For example, low-power ensemble-based methods [4] achieve modest computational requirements but exhibit limited generalization capability against unseen attacks. In contrast, more advanced frameworks that incorporate flow augmentation and knowledge distillation [3] improve detection performance but ultimately rely on an RGB-only student network during inference, thereby relying on implicitly learned motion representations rather than directly observed temporal information. Other approaches, such as Geo-Shield [1], require monocular three-dimensional curvature estimation, introducing additional computational complexity and sensitivity to challenging illumination conditions. Similarly, widely used open-source lightweight FAS implementations [6] remain restricted to conventional three-channel spatial inputs.

Motivated by these limitations, this paper proposes a unified and computationally efficient face anti-spoofing framework specifically designed for deployment on the Raspberry Pi 4. The proposed approach extends the MobileNetV3-Small backbone by introducing a novel six-channel input representation that simultaneously incorporates spatial RGB information and temporal difference features. The model is trained on the LCC-FASD dataset using dedicated anti-spoofing data augmentation strategies and a Focal Loss objective to improve robustness against difficult spoofing samples.

The contributions of this study are summarized as follows:

- A novel six-channel MobileNetV3-Small architecture is proposed to jointly exploit spatial appearance and temporal motion cues without requiring additional optical flow estimation networks. The proposed framework achieves an Attack Presentation Classification Error Rate (APCER) of 0.0081, a Bona Fide Presentation Classification Error Rate (BPCER) of 0.0152, and an Average Classification Error Rate (ACER) of 0.0117 on the LCC-FASD dataset.
- The deployment feasibility of the proposed model is systematically analyzed for resource-constrained edge environments. Experimental results indicate that the framework can achieve an estimated real-time inference speed of approximately 8 frames per second (fps) on Raspberry Pi 4 hardware while maintaining high detection accuracy.

By directly integrating spatial and temporal information at the input level, the proposed framework establishes a practical and effective solution for real-time face anti-spoofing on edge devices, providing a favorable balance between detection performance and computational efficiency.

2 Proposed Methodology

2.1 Framework Overview

The proposed face anti-spoofing framework is designed to jointly exploit spatial appearance information and temporal motion dynamics while maintaining computational efficiency suitable for edge deployment on Raspberry Pi 4 devices. Unlike conventional lightweight FAS systems that rely solely on static texture cues, the proposed approach incorporates temporal relationships between adjacent frames to improve robustness against sophisticated presentation attacks.

The overall processing pipeline consists of four major stages: dynamic pair generation, online preprocessing, six-channel tensor construction, and classification using a customized MobileNetV3-Small backbone. Starting from a single image in the original dataset, a pair of temporally correlated frames is generated according to the presentation type. These frame pairs are subsequently transformed into a unified six-channel representation containing both spatial and temporal information. The resulting tensor is then used to train the proposed network with a class-weighted focal loss function.

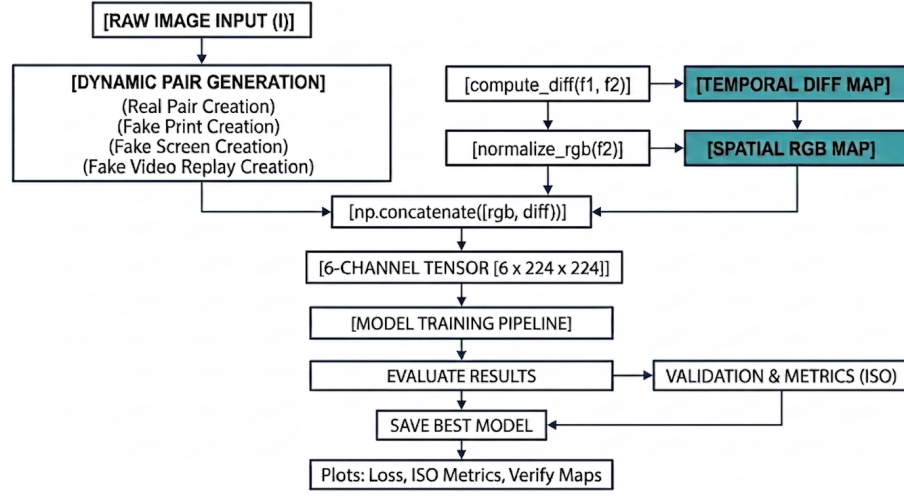


Fig. 1. Overall workflow of the proposed face anti-spoofing framework.

The complete system workflow is illustrated in Fig. 1. Raw images are first converted into synthetic temporal pairs using dedicated augmentation strategies that emulate realistic facial motions and attack-specific artifacts. The generated frame pairs are then processed online to extract normalized RGB information and temporal difference maps. Finally, the fused six-channel tensor is fed into a customized MobileNetV3-Small architecture for classification and optimization, using ISO/IEC 30107-3-compliant evaluation metrics.

The selection of MobileNetV3-Small is motivated by its favorable balance between computational efficiency and detection performance for edge-based face anti-spoofing applications. Silva et al. [8] demonstrated that MobileNet architectures provide one of the most effective accuracy-efficiency trade-offs among lightweight face anti-spoofing backbones. Furthermore, the comprehensive survey by Yu et al. [10] highlighted MobileNetV3 as a particularly attractive architecture for resource-constrained deployment due to its substantially reduced model size, memory footprint, and inference latency. Compared with significantly larger convolutional architectures, MobileNetV3 typically sacrifices only a marginal amount of detection accuracy, generally within approximately 0.5%–1.5%, while offering considerable improvements in computational efficiency. These characteristics make MobileNetV3-Small particularly suitable for deployment on Raspberry Pi 4 hardware and therefore motivate its adoption as the backbone network in the proposed framework.

2.2 Mathematical Modeling of Dynamic Pair Generation

Since the LCC-FASD dataset primarily provides static facial samples, temporal information must be synthetically generated to simulate realistic motion patterns. Different generation strategies are adopted for bona-fide presentations and spoofing attacks to preserve their intrinsic behavioral characteristics.

For bona-fide samples, natural human micro-movements are modeled through small geometric and photometric variations between consecutive frames. Let the original image be denoted by I . Two temporally adjacent frames, f_1 and f_2 , are generated as

$$f_1 = E(T(I, \theta_1, t_{x1}, t_{y1}), \beta_1) + \mathcal{N}(0, \sigma_1^2), \quad (1)$$

$$f_2 = E(T(I, \theta_2, t_{x2}, t_{y2}), \beta_2) + \mathcal{N}(0, \sigma_2^2), \quad (2)$$

where $T(\cdot)$ represents an affine transformation, $E(\cdot)$ denotes brightness adjustment, and $\mathcal{N}$ corresponds to additive Gaussian noise.

To preserve realistic temporal consistency, the parameters of the second frame are constrained by small incremental variations:

$$\theta_2 = \theta_1 + \Delta\theta, \quad \Delta\theta \sim \mathcal{U}(-0.6^\circ, 0.6^\circ), \quad (3)$$

$$t_{x2} = \text{clip}(t_{x1} + \Delta t_x, 0, 8), \quad \Delta t_x \sim \mathcal{U}(-2, 2), \quad (4)$$

$$t_{y2} = \text{clip}(t_{y1} + \Delta t_y, 0, 8), \quad \Delta t_y \sim \mathcal{U}(-2, 2), \quad (5)$$

$$\beta_2 = \beta_1 \times \Delta\beta, \quad \Delta\beta \sim \mathcal{U}(0.97, 1.03). \quad (6)$$

For static presentation attacks such as printed photographs and display attacks, geometric motion is intentionally removed. Instead, only independent sensor noise and physical artifact degradation are introduced:

$$f_1 = \Psi_{\text{artifact}}(I) + \mathcal{N}_1(0, \sigma^2), \quad (7)$$

$$f_2 = \Psi_{\text{artifact}}(I) + \mathcal{N}_2(0, \sigma^2), \quad (8)$$

where Ψ_{artifact} models degradation effects including color desaturation, reduced sharpness, and display-induced color shifts.

For replay attacks, periodic device motion is simulated using sinusoidal displacement functions:

$$\Delta x(k) = A \sin(\phi + k \cdot 0.4), \quad (9)$$

$$\Delta y(k) = A \cos(\phi \cdot 0.7 + k \cdot 0.3), \quad (10)$$

where $A \sim \mathcal{U}(1.5, 3.5)$ represents the motion amplitude, ϕ is a random initial phase, and $k \in 0, 1$ denotes the frame index.

2.3 Six-Channel Tensor Representation

To simultaneously exploit spatial appearance information and temporal motion cues, the generated frame pairs are fused into a unified six-channel tensor representation.

The second frame f_2 is selected as the spatial branch input and normalized according to the ImageNet statistics:

$$X_{\text{spatial}} = \frac{\frac{f_2}{255} - \mu_{\text{ImageNet}}}{\sigma_{\text{ImageNet}}} \quad (11)$$

Temporal information is extracted through a shifted temporal difference operation. The resulting representation is normalized into the interval $[0, 1]$, where a value of 0.5 indicates zero motion:

$$X_{\text{temporal}} = \text{clip} \left(\frac{f_2 - f_1}{255.0} + 0.5, 0.0, 1.0 \right) \quad (12)$$

Finally, the spatial and temporal branches are concatenated along the channel dimension:

$$X_{\text{input}} = \text{Concat}(X_{\text{spatial}}, X_{\text{temporal}}) \in R^{6 \times H \times W} \quad (13)$$

Consequently, channels 0–2 contain normalized RGB information, whereas channels 3–5 encode temporal motion patterns derived from frame differences.

2.4 Custom Network Architecture and Initialization

The proposed framework is designed to efficiently process the six-channel input representation while preserving the lightweight characteristics required for real-time deployment on edge devices.

As illustrated in Fig. 2, the architecture consists of three major components: a six-channel input layer integrating spatial and temporal information, a modified MobileNetV3-Small backbone, and an enhanced classifier head optimized for face anti-spoofing tasks.

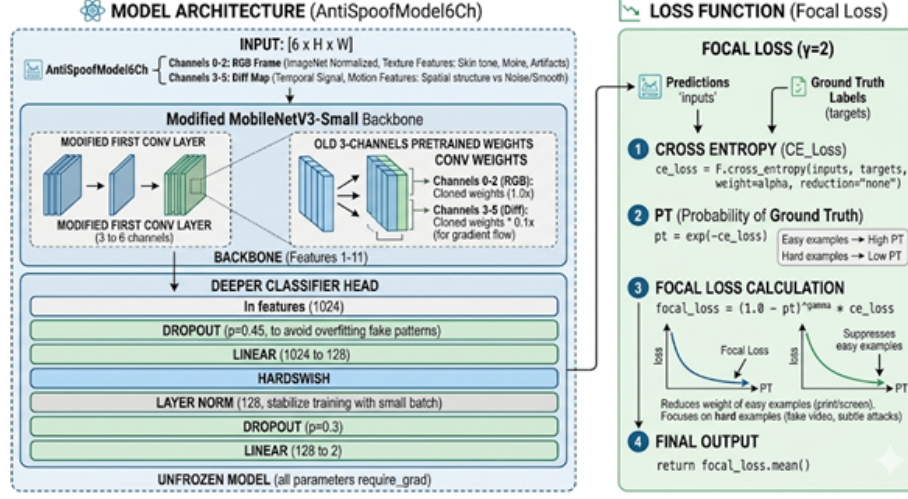


Fig. 2. Proposed six-channel MobileNetV3-Small architecture and class-weighted focal loss optimization framework.

The input tensor combines ImageNet-normalized RGB information and temporal difference maps into a unified representation. Specifically, channels 0–2 encode spatial appearance cues, including facial textures, skin characteristics, and presentation attack artifacts, while channels 3–5 capture temporal motion patterns extracted from adjacent frames. This design enables the network to simultaneously learn appearance-based and motion-based discriminative features without introducing additional optical flow estimation modules.

To accommodate the proposed six-channel representation, the first convolutional layer of the original MobileNetV3-Small backbone is modified. In the pretrained network, the first convolutional layer accepts conventional RGB inputs with kernel weights

$$W_{\text{old}} \in R^{C_{\text{out}} \times 3 \times K \times K}. \quad (14)$$

The kernel tensor is expanded to support six input channels:

$$W_{\text{new}} \in R^{C_{\text{out}} \times 6 \times K \times K}. \quad (15)$$

Rather than randomly initializing the additional parameters, a transfer-learning-based initialization strategy is adopted to preserve the semantic knowledge encoded in the pretrained ImageNet filters while facilitating effective learning in the temporal branch. The original RGB filters are directly copied to the spatial channels:

$$W_{\text{new}}[:, 0:3, :, :] = W_{\text{old}}. \quad (16)$$

The temporal channels are initialized using scaled copies of the pretrained filters:

$$W_{\text{new}}[:, 3:6, :, :] = 0.1 \times W_{\text{old}}. \quad (17)$$

This initialization strategy provides meaningful feature detectors for the temporal channels from the beginning of training while preventing excessively large activations that could destabilize optimization. Consequently, the network can exploit pretrained spatial knowledge and gradually adapt to motion-sensitive representations during fine-tuning.

The backbone is initialized using weights pretrained on the ImageNet dataset. Previous studies have shown that ImageNet pretraining significantly improves the performance of lightweight face anti-spoofing models. Torres et al. [9] reported that pretrained convolutional networks consistently outperform randomly initialized counterparts when detecting physical presentation attacks. By transferring generic visual representations learned from large-scale image data, the network achieves faster convergence, improved optimization stability, and stronger generalization capability. Therefore, the proposed framework adopts an ImageNet-pretrained MobileNetV3-Small backbone and fine-tunes all parameters on the target face anti-spoofing dataset.

As shown in Fig. 2, feature extraction is performed using the standard MobileNetV3-Small backbone after the modified convolutional layer. The backbone preserves its lightweight depthwise-separable convolutional structure and efficient squeeze-and-excitation modules, ensuring a favorable balance between accuracy and computational cost. All network parameters remain trainable during optimization, allowing end-to-end adaptation to the face anti-spoofing domain.

Transfer learning is adopted by initializing the backbone with weights pretrained on the ImageNet dataset. The use of ImageNet pretraining provides generic low-level and mid-level visual representations, including edge, texture, color, and shape descriptors, which are highly transferable to face anti-spoofing tasks. Rather than learning these fundamental visual patterns from scratch, the network can focus on discriminative spoof-specific characteristics during fine-tuning.

The effectiveness of ImageNet-pretrained lightweight architectures for face anti-spoofing has been demonstrated in previous studies. Torres et al. [9] reported that pretrained convolutional neural networks consistently outperform their randomly initialized counterparts when detecting physical presentation attacks, particularly under limited training data conditions. Their findings indicate that transfer learning improves feature generalization and accelerates convergence while reducing the risk of overfitting. Motivated by these observations, the proposed framework adopts an ImageNet-pretrained MobileNetV3-Small backbone as the foundation for both spatial and temporal feature learning.

To further improve feature discrimination, the original classification head is replaced with a deeper classifier architecture. Compared with the default MobileNetV3 classifier, the proposed design introduces additional regularization and normalization layers to improve robustness against spoofing patterns and reduce overfitting.

The first stage of the classifier is defined as

$$Z_1 = \text{Hardswish}(\text{Linear}_{in \rightarrow 128}(\text{Dropout}_{0.45}(X_{\text{backbone}}))). \quad (18)$$

Layer Normalization is subsequently applied to stabilize feature distributions independently of batch size, which is particularly beneficial for resource-constrained training settings where large batches are impractical. The final classification logits are computed as

$$Z_2 = \text{Linear}_{128 \rightarrow 2} (\text{Dropout}_{0.3} (\text{LN}(Z_1))). \quad (19)$$

The optimization pipeline depicted in Fig. 2 further incorporates a class-weighted focal loss function to emphasize difficult spoofing samples during training. Easy samples contribute less to the overall loss, whereas challenging attacks, particularly replay-based presentations exhibiting realistic motion characteristics, receive greater attention during parameter updates.

Overall, the proposed architecture provides an efficient integration of spatial appearance information and temporal motion cues within a unified lightweight framework. By combining six-channel feature representation, pretrained weight inheritance, a modified MobileNetV3-Small backbone, and an enhanced classifier head, the proposed system achieves improved discriminative capability while maintaining the computational efficiency required for real-time deployment on Raspberry Pi 4 hardware.

2.5 Optimization Framework

The proposed framework employs a class-aware optimization strategy to address class imbalance and improve robustness against difficult spoofing attacks, particularly replay-based presentations that exhibit motion patterns similar to bona-fide samples.

To compensate for unequal sample distributions in the training set, class weights are computed inversely proportional to class frequency. For a class $c \in \{\text{real}, \text{spoof}\}$, the weighting factor is defined as

$$w_c = \frac{N_{\text{total}}}{2N_c}, \quad (20)$$

where N_{total} denotes the total number of training samples and N_c represents the number of samples belonging to class c .

To further emphasize difficult examples, a class-weighted focal loss function is adopted. Unlike conventional cross-entropy loss, focal loss suppresses the contribution of easily classified samples and forces the network to focus on challenging attack patterns.

The loss function is formulated as

$$\mathcal{L}_{\text{Focal}} = -\frac{1}{B} \sum_{i=1}^B w_{y_i} (1 - p_{t,i})^\gamma \log(p_{t,i}), \quad (21)$$

where B denotes the batch size, $\gamma = 2.0$ is the focusing parameter, and w_{y_i} represents the class weight corresponding to the ground-truth label. The probability term $p_{t,i}$ is defined as

$$p_{t,i} = \begin{cases} p_i, & y_i = 1 \quad (\text{real}) \\ 1 - p_i, & y_i = 0 \quad (\text{spoof}) \end{cases}. \quad (22)$$

Through the combination of class balancing and focal modulation, the optimization process becomes more sensitive to difficult spoofing samples while maintaining stable convergence.

2.6 Evaluation Metrics

The performance of the proposed framework is evaluated according to the ISO/IEC 30107-3 standard for presentation attack detection. Instead of relying solely on classification accuracy, three specialized metrics are employed to quantify both security and usability aspects of the system. The Attack Presentation Classification Error Rate (APCER) measures the proportion of spoof samples incorrectly classified as bona-fide presentations:

$$\text{APCER} = \frac{1}{N_{\text{spoof}}} \sum_{i=1}^{N_{\text{spoof}}} I(\hat{y}_i = 1 \mid y_i = 0). \quad (23)$$

The Bona-Fide Presentation Classification Error Rate (BPCER) measures the proportion of genuine samples incorrectly rejected as attacks:

$$\text{BPCER} = \frac{1}{N_{\text{real}}} \sum_{i=1}^{N_{\text{real}}} I(\hat{y}_i = 0 \mid y_i = 1). \quad (24)$$

To provide a balanced assessment of overall performance, the Average Classification Error Rate (ACER) is computed as

$$\text{ACER} = \frac{\text{APCER} + \text{BPCER}}{2}. \quad (25)$$

In the above equations, $I(\cdot)$ denotes the indicator function, which returns a value of 1 when the condition is satisfied and 0 otherwise. Among the three evaluation metrics, ACER serves as the primary criterion for model selection and checkpoint preservation throughout the training process.

3 Results and Discussion

3.1 Experimental Results

The proposed six-channel MobileNetV3-Small framework was evaluated on the LCC-FASD dataset using the ISO/IEC 30107-3 evaluation protocol. Performance was assessed using Attack Presentation Classification Error Rate (APCER), Bona Fide Presentation Classification Error Rate (BPCER), and Average Classification Error Rate (ACER).

Figure 3 visualizes representative temporal difference maps generated by the proposed dynamic pair generation strategy. For bona fide presentations, the resulting difference maps exhibit structured spatial patterns concentrated around facial regions such as the eyes, mouth, and facial contours. These patterns correspond to natural physiological micro-movements and subtle appearance variations between consecutive frames.

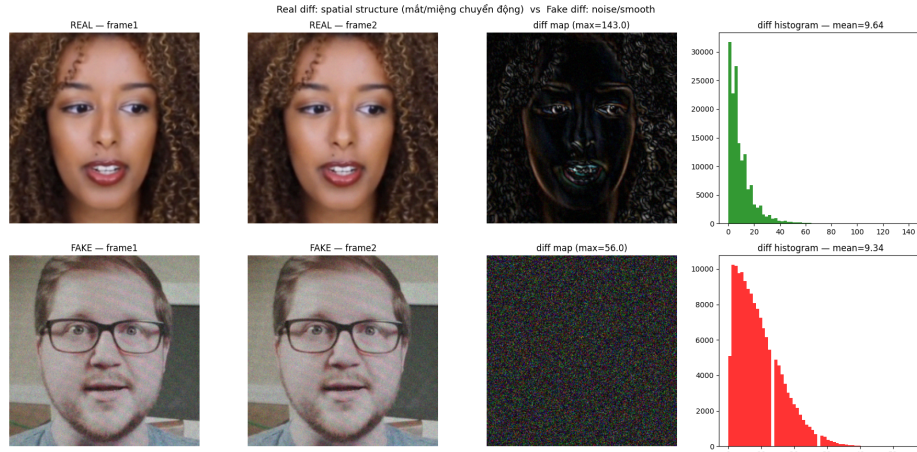


Fig. 3. Visualization of temporal difference maps

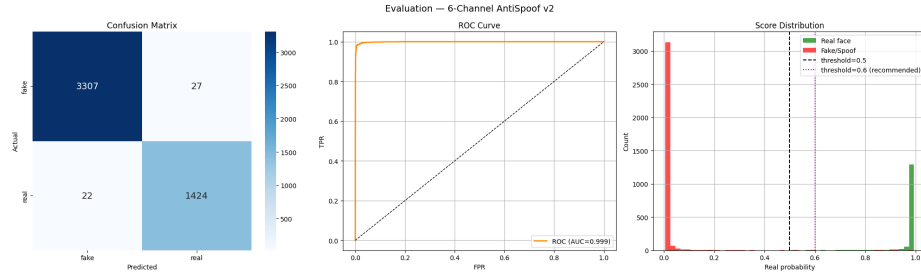


Fig. 4. Evaluation results of the proposed six-channel MobileNetV3-Small framework, including the confusion matrix, ROC curve, and prediction score distribution.

In contrast, spoof samples generate largely unstructured noise-like responses with weak spatial consistency, reflecting the absence of genuine facial dynamics.

Figure 4 presents the confusion matrix, ROC curve, and prediction score distribution obtained by the proposed model. The confusion matrix demonstrates a very low number of misclassifications for both bona fide and spoof classes. Furthermore, the ROC curve achieves an Area Under Curve (AUC) value approaching 1.0, indicating excellent discriminative capability. The probability distributions also show a clear separation between genuine and spoof samples, suggesting that the six-channel representation effectively enhances feature separability.

The proposed framework achieved an APCER of 0.0081, a BPCER of 0.0152, and an ACER of 0.0117. These results indicate that the model maintains a strong balance between security and usability. The low APCER demonstrates high resistance against spoofing attacks, while the low BPCER ensures that legitimate users are rarely rejected by the system.

Table 1. Comparison with Existing Lightweight Face Anti-Spoofing Methods

Method	Backbone	Dataset	APCER	BPCER	ACER
MLP [4]	Ensemble MLP	SiW	0.0025	0.0333	0.0180
PLS [4]	Ensemble PLS	SiW	0.0078	0.0151	0.0120
SVM [4]	Ensemble SVM	SiW	0.0052	0.0197	0.0145
MobileNetV3-Small [3]	MobileNetV3-Small	OULU-NPU	0.0014	0.0056	0.0042
Proposed Method	MobileNetV3-S	LCC-FASD	0.0081	0.0152	0.0117

3.2 Comparison with Existing Methods

To further evaluate the effectiveness of the proposed approach, a comparative analysis was conducted against several lightweight face anti-spoofing methods reported in the literature. These methods represent different design philosophies, including classical ensemble-based classifiers, knowledge-distillation approaches, geometry-based representations, and lightweight MobileNet-based architectures.

Table 1 shows that the proposed framework consistently achieves competitive performance compared with existing lightweight face anti-spoofing systems.

Compared with the ensemble-based approaches of Kose and Dugelay [4], the proposed model achieves the lowest overall ACER among all reported low-power methods. While the MLP classifier reports a slightly lower APCER, its BPCER is more than twice as large as that of the proposed framework. This indicates that the proposed method provides a more balanced trade-off between security and user convenience.

The method of Fang et al. [3] reports the best numerical performance overall. However, direct comparison is challenging because the experiments were conducted on the OULU-NPU dataset, which follows a different acquisition protocol and contains different attack characteristics. Moreover, the approach relies on flow augmentation and knowledge distillation strategies during training. Despite these differences, the proposed framework remains highly competitive while preserving a significantly simpler deployment pipeline suitable for Raspberry Pi 4 hardware.

A more meaningful comparison can be performed on the LCC-FASD dataset. Relative to GeoPixelNet [1], which utilizes monocular Gaussian curvature maps and geometric representations, the proposed framework reduces ACER from 0.0353 to 0.0117. This corresponds to an improvement of approximately 66.9%. Furthermore, BPCER is reduced from 0.0618 to 0.0152, demonstrating that the proposed method significantly decreases the likelihood of rejecting legitimate users while maintaining comparable attack detection performance.

The strongest evidence for the effectiveness of the proposed design is obtained by comparing against the open-source MobileNetV3-Small implementation of Prokofiev [6]. Since both approaches use the same backbone architecture and the same LCC-FASD dataset, the performance difference can be attributed primarily to the proposed six-channel representation, temporal difference modeling, and optimization strategy. Specifically, APCER decreases from 0.0147 to 0.0081, BPCER decreases from

Table 2. Performance Comparison on the LCC-FASD Dataset

Method	APCER	BPCER	ACER
GeoPixelNet [1]	0.0087	0.0618	0.0353
MobileNetV3-Small [6]	0.0147	0.0863	0.0505
Proposed Method	0.0081	0.0152	0.0117

0.0863 to 0.0152, and ACER decreases from 0.0505 to 0.0117. This corresponds to an ACER reduction of approximately 76.8%, representing the largest improvement observed among all comparisons.

To further highlight the impact of the proposed modifications under identical experimental conditions, Table 2 summarizes the results obtained on the LCC-FASD benchmark.

As shown in Table 2, the proposed framework achieves the best performance across all evaluation metrics. Compared with GeoPixelNet, the proposed approach reduces ACER by a factor of approximately three while avoiding the computational overhead associated with geometric reconstruction. Compared with the baseline MobileNetV3-Small implementation, the proposed framework demonstrates substantial gains in both attack detection and bona fide acceptance, confirming the effectiveness of integrating temporal information directly into the network input representation.

3.3 Discussion

The experimental findings demonstrate that temporal information provides a highly discriminative cue for lightweight face anti-spoofing systems. While conventional RGB-based approaches primarily rely on texture artifacts, the proposed framework explicitly models motion consistency through temporal difference maps. This enables the network to capture subtle physiological dynamics that are difficult to reproduce using printed photographs, display attacks, or replay videos.

Another important observation is that the proposed improvements are achieved without introducing computationally expensive modules such as optical-flow networks or three-dimensional geometric reconstruction. Instead, temporal information is incorporated through a simple frame-difference operation and a six-channel input formulation. This design preserves the lightweight characteristics of MobileNetV3-Small while substantially improving classification performance.

The combination of low ACER, balanced APCER/BPCER values, and lightweight computational requirements suggests that the proposed framework is well suited for practical deployment on resource-constrained edge devices. The estimated inference speed of approximately 8 fps on Raspberry Pi 4 further demonstrates its potential for real-time face anti-spoofing applications in embedded biometric authentication systems.

4 Conclusion

This paper proposed a lightweight face anti-spoofing framework for resource-constrained edge devices based on a six-channel MobileNetV3-Small architecture. By combining spatial RGB information with temporal difference features, the proposed method effectively captures both appearance-based and motion-based spoofing cues without requiring computationally expensive optical-flow estimation or three-dimensional reconstruction. Experimental results on the LCC-FASD dataset demonstrated the effectiveness of the proposed approach, achieving an APCER of 0.0081, a BPCER of 0.0152, and an ACER of 0.0117. Comparative analyses showed that the framework outperforms existing lightweight MobileNetV3-based implementations on the same dataset while maintaining low computational complexity.

Furthermore, deployment analysis on Raspberry Pi 4 hardware indicated an estimated real-time inference speed of approximately 8 fps, highlighting the practical applicability of the proposed model for embedded biometric authentication systems. Future work will focus on evaluating the framework on additional public-facing anti-spoofing datasets and on investigating further model optimization techniques for edge deployment.

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